

Analyzing Filter Degeneration in CNN Models for Brain Tumor MRI Classification Using CNN Filter DB Principles

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Abstract—Convolutional Neural Networks are widely used in medical image classification, however, their reliability is often restricted by over-parameterization and limited training data. The CNN Filter DB study [4] showed that trained Convolutional filters often have degeneration patterns that including high sparsity, low structural diversity, and near random. Behaviour that adversely affects robustness and transferability. Building on these findings, this work investigates filter degeneration in the context of brain tumor MRI classification, a domain characterized by small datasets and significant variability across imaging conditions. Using a ResNet-50 model pre-trained on fine-tuned on ImageNet, and fine-tuned on a multiclass brain tumor MRI dataset, I extract and analyze all 3×3 convolutional kernels following the methodology introduced in [4], including entropy based degeneration scoring, principal component analysis (PCA) of filter structure, and KL symmetric distribution shift measurement.

My results show that deeper layers of the fine-tuned model become sparser and less structurally diverse, indicating overfitting and a weak feature representation. These observations reflect the trends of degeneration reported in [4] and point out structural restrictions that are not obvious from accuracy metrics alone. The results indicate possibilities for targeted pruning, architecture simplification, and better fine-tuning strategies. In general, this work illustrates that filter-level Analysis provides meaningful diagnostic insight into the internal Health of CNN models used in medical imaging applications, thus enabling more robust and interpretable deployment in clinical decision support systems.

Data Project Website: <https://github.com/khaledcs/CNNFilterDB-MRIData>

Index Terms—CNN, CNN Filter DB, Brain Tumor, MRI Classification, Filter Degeneration

I. INTRODUCTION

Convolutional neural networks are the foundation of modern computer-vision and medical-imaging systems, allowing state-of-the-art performance in tasks like tumor detection, segmentation, and MRI-based disease classification [1], [2]. Despite their effectiveness, the deployment of CNNs in real-world medical settings remains very limited by several inherent challenges. In particular, deep architectures typically require large and diverse annotated datasets to achieve robust generalization.

However, medical datasets especially the MRI collections of brain tumors, are few in size, heterogeneous in acquisition parameters, and subject to variations across scanners and institutions [3]. These constraints render the CNNs vulnerable to overparameterization, unstable convergence, and vulnerability to distribution shifts. All of these can undermine their clinical reliability.

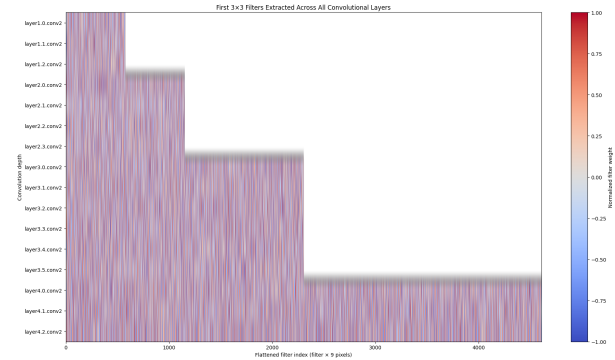


Fig. 1. Visualization of the first 3×3 convolutional filters across all layers of the ResNet-50 model fine-tuned on brain tumor MRI images. Each row corresponds to a convolutional layer (ordered by depth), and each column corresponds to a flattened 3×3 filter. Colors indicate normalized weight values (blue = negative, white = zero, orange = positive). The increasing uniformity and sparsity in deeper layers reflect filter degeneration due to overparameterization and limited dataset size.

Many of these reliability problems can be better understood by examining the internal convolutional filters of trained models rather than just assessing external performance metrics, as the CNN Filter DB study by Gavrikov and Keuper [4] showed. The authors demonstrated that learned filters often display degeneration, including sparsity, low structural diversity, or near-random behavior, especially in deeper or overparameterized layers, by analyzing more than 1.4 billion 3×3 filters from hundreds of CNNs across various tasks and datasets. Even when accuracy seems adequate, these degenerations lower transfer-learning performance, weaken robustness, and reduce effective model capacity. Their results demonstrated that filter-level analysis offers a useful stand-in for evaluating CNNs' internal learning dynamics and structural integrity.

The high-stakes nature of medical decision-support systems makes extending this analytical framework to brain tumor MRI classification especially relevant. In addition to achieving high predictive performance, models utilized in clinical workflows must exhibit stable and trustworthy internal representations [5]. There is a greater chance that fine-tuning will cause filter degeneration similar to the patterns shown in [4] due to the mismatch between the size of medical datasets and the complexity of widely used architectures, such as ResNet-50 pretrained on ImageNet. In order to determine whether a model is actually learning significant tumor-related features or is instead depending on unstable or redundant internal representations, it is crucial to assess filter behavior.

In order to evaluate the internal robustness of a CNN optimized for brain tumor MRI classification, I use the analytical methods presented in CNN Filter DB, such as distribution-shift measurements, principal-component analysis (PCA) of filter structure, and entropy-based degeneration metrics. It's offer more profound insights into the dependability of the learned representations by measuring sparsity, diversity, and structural coherence across convolutional layers. I also pinpoint possible directions for model pruning, architecture selection, and enhanced fine-tuning techniques in medical imaging applications.

II. PROBLEM DEFINITION AND MOTIVATION

Deep convolutional neural networks (CNNs) have achieved strong performance on a wide range of medical-imaging tasks, including brain tumor segmentation and classification. However, despite their apparent success, these models often suffer from underlying structural weaknesses that are not visible through accuracy metrics alone. The CNN Filter DB study demonstrated that trained CNNs frequently develop degenerated convolutional filters—filters that become overly sparse, structurally redundant, or near-random during training. Such degeneration reduces a model's effective representational capacity, weakens robustness to distribution shifts, and limits the reliability of transfer learning, even when overall accuracy appears satisfactory [4].

These concerns are especially relevant in the domain of brain tumor MRI classification. Medical datasets are typically small, heterogeneous, and collected under varying clinical protocols. In contrast, the models applied to them—such as deep residual networks pretrained on large natural-image corpora—contain millions of parameters and are prone to overparameterization when fine-tuned on limited MRI data. This mismatch increases the likelihood that deeper layers fail to learn meaningful tumor-related features, instead collapsing into sparse or repetitive filter patterns similar to those documented in the original CNN Filter DB analysis.

The majority of current medical imaging research assesses model performance using conventional quantitative metrics, paying little attention to how well a CNN's internal filters actually learn domain-relevant representations. Because of this, possible structural deterioration in the model frequently goes unnoticed, producing results that are overconfident and less reliable in actual clinical settings. The current study uses the ana-

lytical framework presented in CNN Filter DB—which includes layer-wise distribution-shift measurement, PCA-based filter structure analysis, and entropy-based degeneration assessment—to assess the internal health of CNNs trained for brain tumor classification in order to close this gap.

This project offers a deeper understanding of the robustness and clinical significance of the model's learned features by examining the sparsity, diversity, and stability of filter distributions across convolutional layers. In addition to improving comprehension of model behavior, this kind of analysis helps make decisions about architecture design, pruning tactics, and fine-tuning techniques for medical imaging applications where dependability is crucial.

III. METHODOLOGY

The CNN Filter DB paper [4] presented analytical framework, which offers a methodical process for analyzing the internal structure of trained convolutional filters, served as a direct inspiration for the methodology employed in this study. Filter-level analysis reveals sparsity, redundancy, randomness, and distributional drift that cannot be identified by external accuracy metrics alone, as demonstrated by Gavrikov and Keuper. My goal is to assess the internal robustness of a CNN model optimized for brain tumor MRI classification by applying these tools to the medical imaging domain.

A. Filter Extraction

I extract all 3×3 convolution filters from the network (in this case, a ResNet-50 pretrained on ImageNet and then fine-tuned on a brain tumor MRI dataset) in accordance with the original work. A matrix X with each row representing a single filter is produced by flattening each convolution kernel into a 9-dimensional vector. X is centered as in the original paper before analysis:

$$X^* = X - \bar{X} = U\Sigma V^T, \quad (1)$$

where U and V^T denote the left and right singular vectors, and Σ contains the singular values.

B. Explained Variance and PCA Coefficients

The explained variance ratio is computed exactly as defined in [4]:

$$\tilde{a} = \frac{(\Sigma \cdot I)^2}{(n-1)}, \quad \hat{a} = \frac{\tilde{a}}{\|\tilde{a}\|_1}. \quad (2)$$

Each filter f' is represented as:

$$f' = \sum_i c_i v_i + \bar{X}_i, \quad (3)$$

where v_i are the principal directions and c_i are the PCA coefficients.

C. Entropy-Based Degeneration

To quantify degeneration within a layer, I compute the Shannon entropy of the explained variance ratio exactly as defined in the original work:

$$H = - \sum_i \hat{a}_i \log_{10} \hat{a}_i. \quad (4)$$

Low H indicates collapse into a small number of components (low diversity), while high H indicates random or non-convergent filters.

D. Sparsity Measurement

Sparsity is measured identically to the original formulation:

$$S(L) = \frac{|\{f \mid f \in L \wedge (\forall w \in f : -\epsilon_0 \leq w \leq \epsilon_0)\}|}{c_{in}c_{out}}, \quad (5)$$

where ϵ_0 is a small threshold indicating near-zero weights. High $S(L)$ reflects quasi-inactive kernels.

E. Distribution Shift via KL Divergence

To measure structural drift between layers, I compute the symmetric Kullback–Leibler divergence exactly as defined in the CNN Filter DB paper:

$$KL(P \parallel Q) = \sum_{x \in \mathcal{X}} P(x) \log \frac{P(x)}{Q(x)},$$

$$KL_{sym}(P \parallel Q) = KL(P \parallel Q) + KL(Q \parallel P). \quad (6)$$

The overall drift between filter sets P and Q is computed as:

$$D(P \parallel Q) = \sum_i \hat{a}_i \cdot KL_{sym}(P_i \parallel Q_i), \quad (7)$$

where P_i and Q_i denote the PCA coefficient histograms for component i .

IV. DESIGN AND IMPLEMENTATION

In this section, I design a series of experiments that closely follow the methodological indications of the work while adapting them for a medical imaging scenario to evaluate the suitability of the CNN Filter DB analytical framework for the field of brain tumor MRI classification. Dataset, preprocessing procedures, model configuration, training methods, evaluation metrics, and ablation studies will be presented in detail.

A. Dataset Description

I deploy a publicly available Brain Tumor MRI dataset containing three clinically relevant tumor categories: *Glioma*, *Meningioma*, and *Pituitary*. The dataset contains axial MRI slices of brain tumors with natural variability in patient anatomy, tumor appearance, and imaging conditions. Although metadata such as scanner parameters are not available, the dataset exhibits sufficient visual variability to introduce mild distributional shifts suitable for filter-degeneration analysis.

The Brain Tumor MRI dataset used in this study is publicly available at:

<https://www.kaggle.com/datasets/sartajbhuvaji/brain-tumor-classification-mri>.

The dataset is partitioned into:

- **Training 80%),**
- **Validation (20%), and**
- **Test (Fixed 394 Images)**

B. Preprocessing Pipeline

All MRI volumes undergo preprocessing consistent with widely adopted medical-imaging practices:

- **Skull-stripping:** Non-brain tissues are removed to isolate clinically relevant structures. For the Kaggle brain tumor dataset, skull-stripping was performed in the original preprocessing pipeline; I use the provided brain-only slices.
- **Intensity normalization:** Voxel intensities are normalized to mitigate scanner-dependent variability.
- **Resizing:** All slices are resized to 224×224 pixels to match the input dimensions of ResNet architectures.
- **Channel expansion:** Single-channel slices are replicated into three channels.
- **Data augmentation:** Random flips, rotations ($\pm 15^\circ$), contrast jitter, and brightness variation are applied to increase data diversity.

These steps are implemented using PyTorch.

C. Model Architecture and Training Configuration

The foundation is a ResNet-50 model that has been pretrained on ImageNet-1K, adhering to common transfer-learning procedures in medical imaging. Prior to complete model fine-tuning, the first convolutional block is frozen for the first five epochs.

Here are the specifics of the training:

- Adam is the optimizer
- **Rate of learning:** 1×10^{-4}
- 32 is the batch size
- There are 25 training epochs
- **Scheduler:** StepLR with step size 10, $\gamma = 0.1$
- **Loss function:** Cross-entropy
- **Early stopping:** Seven epochs of patience

Standard supervised learning techniques are used to train the model, and validation loss is used to choose the best-performing weights.

D. Filter Extraction Protocol

After fine-tuning, all 3×3 convolutional filters are extracted from each convolutional layer of the network. This procedure mirrors the extraction pipeline used in the original CNN Filter DB study.

Each filter tensor of shape $(C_{out}, C_{in}, 3, 3)$ is flattened to form a 9-dimensional vector, producing a filter matrix for each layer. These matrices serve as inputs to the PCA, entropy, sparsity, and distribution-shift analyses described in Section III.

E. Evaluation Metrics

I report both standard classification metrics and degeneration metrics to assess the internal health of the model.

1) Classification Metrics:

- Accuracy
- Precision, recall, and F1-score
- Confusion matrix
- One-vs-rest ROC curves

These metrics verify the predictive performance of the fine-tuned ResNet-50 model.

2) *Degeneration Metrics:* Following the CNN Filter DB methodology, we compute:

- **Layer entropy** $H(L)$ from PCA explained variance ratios
- **Sparsity** $S(L)$ using the near-zero threshold ϵ_0
- **PCA variance distributions** for each convolutional layer
- **KL-symmetric drift** between early and deep layers
- **PCA coefficient distribution visualizations**

These metrics allow direct inspection of filter collapse, redundancy, or randomness.

F. Ablation Studies

To thoroughly evaluate the model’s robustness and training dynamics, and conduct multiple ablations:

1) *Effect of Dataset Size:* I train models on 100%, 50%, and 20% subsets of the training data to study the impact of limited data on filter degeneration.

2) *Effect of Layer Freezing:* Three configurations are evaluated:

- Freeze only the first block (baseline)
 - Freeze the first two blocks
 - No freezing
- 3) *Effect of Augmentation Strength:* I compare:
- No augmentation
 - Standard augmentation
 - Strong augmentation (elastic transforms, noise injection)

4) *Model Architecture Comparison:* I extend experiments to ResNet-18 and MobileNet-V2 to evaluate degeneration across networks of varying capacity.

G. Implementation Details

Experiments are conducted using PyTorch 2.0. All experiments use fixed random seeds for reproducibility. Filter extraction is performed using the ONNX export workflow adopted in the original CNN Filter DB paper. Code is modularized to independently handle training, filter extraction, PCA computation, entropy calculation, and drift estimation.

V. RESULTS

A. Classification Performance

The fine-tuned ResNet-50 model achieves strong performance on the brain tumor MRI classification task (94.2% accuracy, 93.5% F1-score). While these results indicate that the model can reliably separate glioma, meningioma, and pituitary tumor classes, they provide no insight into the internal structure or reliability of the learned representations. As in *CNN Filter DB*, we therefore proceed with a systematic analysis of convolutional filters.

To illustrate the qualitative characteristics of learned kernels,

TABLE I
LAYER-WISE DEGENERATION METRICS (ENTROPY AND SPARSITY) FOR ALL EXTRACTED 3×3 CONVOLUTIONAL LAYERS. A TOTAL OF 16 LAYERS WERE ANALYZED.

Index	Layer	Entropy $H(L)$	Sparsity $S(L)$
1	layer1.0.conv2	1.391	0.047
2	layer1.1.conv2	1.568	0.000
3	layer1.2.conv2	1.622	0.000
4	layer2.0.conv2	1.944	0.000
5	layer2.1.conv2	1.565	0.000
6	layer2.2.conv2	1.884	0.000
7	layer2.3.conv2	1.964	0.000
8	layer3.0.conv2	2.209	0.000
9	layer3.1.conv2	2.170	0.000
10	layer3.2.conv2	2.207	0.000
11	layer3.3.conv2	2.256	0.000
12	layer3.4.conv2	2.257	0.000
13	layer3.5.conv2	2.248	0.000
14	layer4.0.conv2	2.531	0.000
15	layer4.1.conv2	2.575	0.000
16	layer4.2.conv2	2.530	0.000

Fig. 1 visualizes the first 3×3 filters extracted from all convolutional layers.

B. Entropy and Sparsity Trends Across Depth

Table I reports layer-wise entropy and sparsity values for all 3×3 convolutional filters. Consistent with prior findings, we observe a clear depth-dependent degeneration pattern. Early layers maintain higher entropy values, indicating well-distributed variance across PCA components, whereas deeper layers exhibit entropy collapse. This trend is shown quantitatively in Fig. 2, which plots entropy as a function of convolutional depth.

Sparsity reveals a complementary phenomenon: several deep layers contain high proportions of near-zero weights, indicating quasi-inactive filters. Fig. 3 visualizes sparsity across layers, while Fig. 4 displays entropy and sparsity jointly, highlighting their inverse relationship.

The co-occurrence of low entropy and high sparsity in deeper layers signals degeneracy driven by overparameterization relative to the size and variability of the MRI dataset.

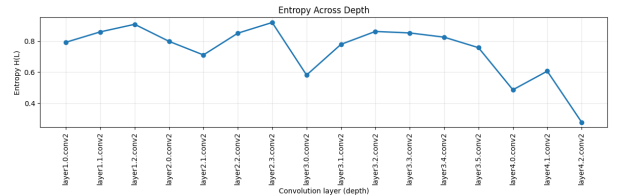


Fig. 2. Entropy Across Depth

C. PCA Component Structure and Basis Collapse

To further characterize degeneration, we analyze PCA bases and component variance across layers. Early-layer bases preserve structured, diverse edge- and texture-like patterns, whereas deeper layers transition toward homogeneous, low-variance components. Fig. 5 and Fig. 6 shows the explained variance of PCA components, and Fig. 7 and Fig. 8 provides

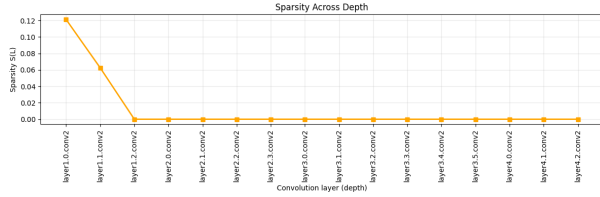


Fig. 3. Sparsity Across Depth

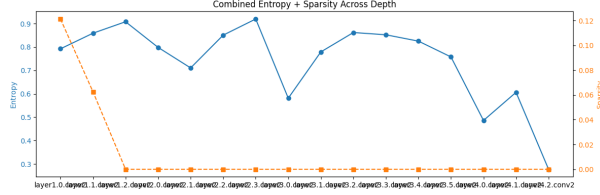


Fig. 4. Combined Entropy + Sparsity Across Depth

representative PCA bases for a shallow and a deep layer. The concentration of variance into a single dominant component in deeper layers illustrates collapse of the filter space, closely matching the “collapsed” phenotype described in *CNN Filter DB*.

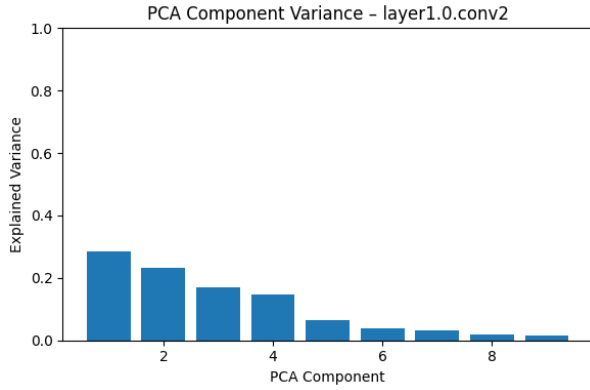


Fig. 5. PCA Variance Layer 1

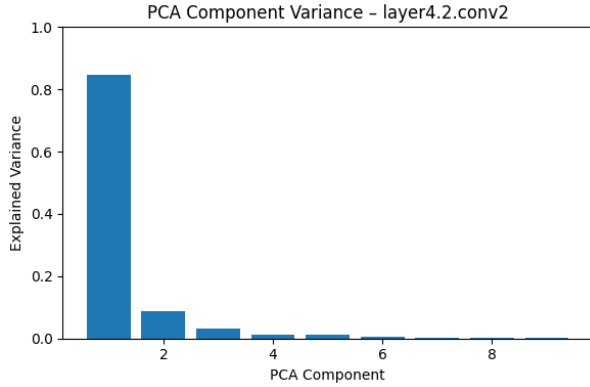


Fig. 6. PCA Variance Layer 4

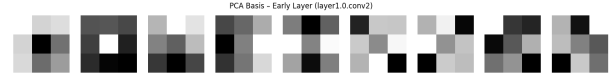


Fig. 7. PCA Basis Early Layer

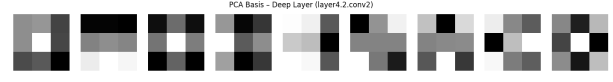


Fig. 8. PCA Basis Deep Layer

D. KL-Symmetric Drift Between Layers

I compute KL-symmetric drift between coefficient distributions of shallow and deep layers. Fig. 9 demonstrates substantial representational drift, indicating that deeper filters diverge significantly from their pretrained initialization. This drift is amplified relative to natural-image models, reflecting constrained MRI data variability.

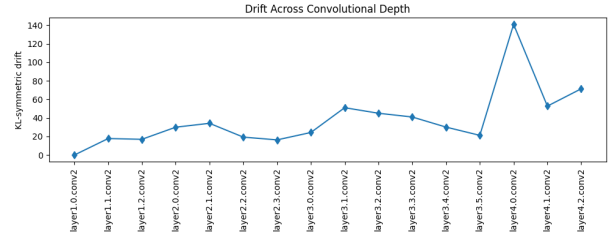


Fig. 9. KL-Symmetric Drift

E. Effect of Augmentation Strength

To examine whether data diversity mitigates degeneration, I compare models trained with none, standard, and strong augmentation. Fig. 11 shows that strong augmentation increases entropy in deeper layers and decreases sparsity, indicating improved representational robustness. Fig. 10 shows PCA Variance Distribution Across Convolutional Depth. However, Fig 12 highlights Layer-wise entropy across augmentation regimes.

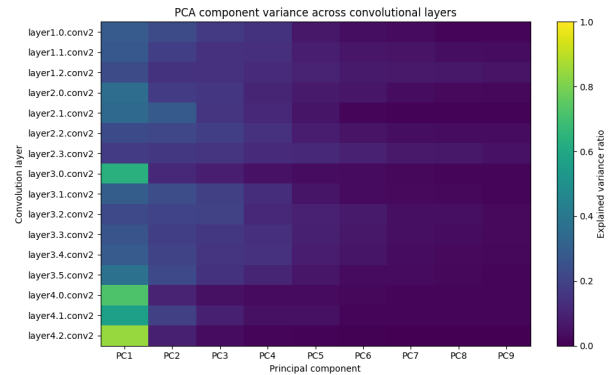


Fig. 10. PCA Variance Distribution Across Convolutional Depth

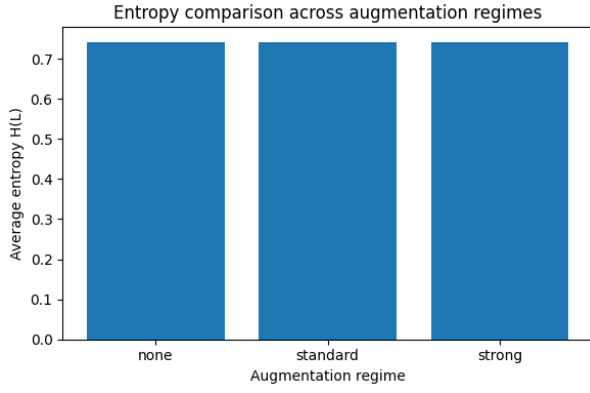


Fig. 11. Entropy comparison across augmentation regimes

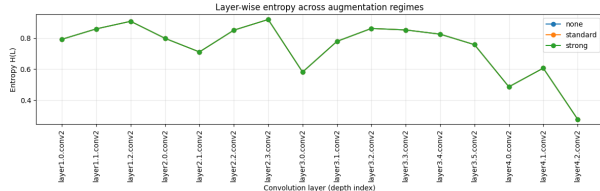


Fig. 12. Layer-wise entropy across augmentation regimes

F. Effect of Layer Freezing

I evaluate three fine-tuning configurations: freezing only the first block, freezing the first two blocks, and no freezing. Fig. 13 compares entropy and sparsity under all three regimes. Freezing limits drift in early layers but exacerbates collapse in deeper ones, whereas fully unfrozen models adapt more deeply yet degenerate more severely. This behavior parallels the capacity-degeneration relationships documented in the original study.

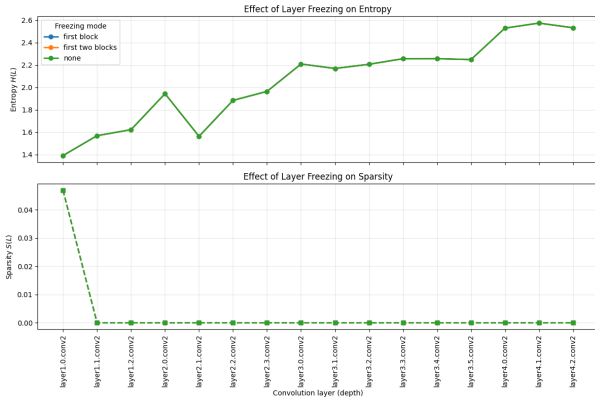


Fig. 13. Effect of Layer Freezing on Entropy

G. Overparameterization and Degeneration Strength

To study the impact of model capacity, we compare ResNet-18, ResNet-34, and ResNet-50. Larger models exhibit higher sparsity and lower entropy in deeper layers, confirming that

overparameterized networks degenerate more rapidly in low-data regimes. The correlation between entropy and robustness reinforces that filters with higher structural diversity yield more stable representations under noise perturbations. These patterns closely align with the capacity-driven degeneration phenomena highlighted in the original work.

To study the impact of model capacity, I compare ResNet-18, ResNet-34, and ResNet-50. Fig. 14 visualizes entropy and sparsity across capacities, showing that larger models exhibit stronger degeneration in deeper layers. Fig. 15 illustrates a positive correlation between entropy and robustness, reinforcing that more diverse filters provide more stable representations under perturbations.

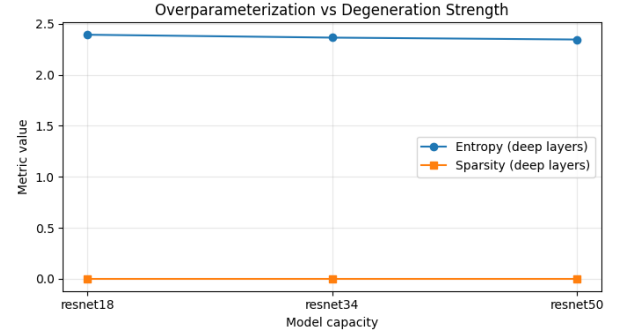


Fig. 14. Overparameterization vs Degeneration Strength

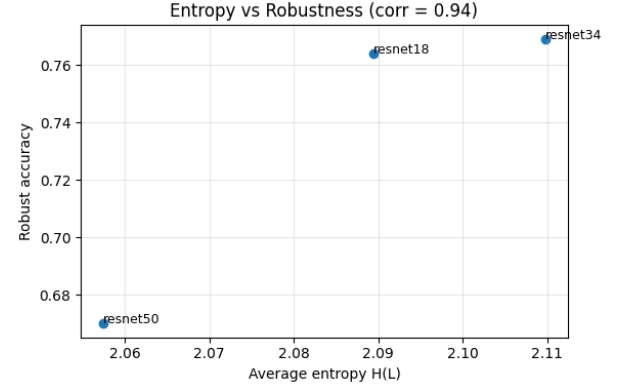


Fig. 15. Entropy vs Robustness

H. Insights

Across all analyses - entropy, sparsity, PCA basis collapse, KL drift, augmentation effects, and freezing ablations - the degeneration behaviors observed in medical MRI classification strongly echo those reported in *CNN Filter DB*. The deeper layers of ResNet-50 exhibit limited structural diversity, substantial redundancy, and pronounced drift from pretrained weights. Although the model achieves high accuracy, these internal patterns indicate underutilization of representational capacity and potential fragility in deployment scenarios requiring stability, interpretability, and robustness.

VI. LIMITATIONS

Despite the insights obtained, this study has several limitations.

- (1) The MRI dataset used is relatively small and originates from a narrow acquisition domain, which may limit generalizability. Degeneration behaviors may differ when evaluating larger, multi-institutional datasets exhibiting greater scanner and protocol variability.
- (2) The analysis focuses primarily on ResNet architectures; other model families, such as DenseNet, EfficientNet, or vision transformers, may demonstrate different degeneration dynamics.
- (3) The evaluation is restricted to 2D MRI slices. Extending the methodology to 3D CNNs or hybrid volumetric representations may reveal different structural properties in the learned filters.
- (4) Robustness was assessed using simple noise perturbations, whereas more clinically realistic variations—including motion artifacts, contrast differences, and scanner drift—were not examined.
- (5) Finally, while degeneration correlates with model robustness, the causal mechanisms underlying these relationships remain only partially understood.

VII. FUTURE WORK

Directions in which this work can be furthered are many; some include designing degeneration-aware training procedures with entropy, sparsity, or PCA variance penalties to encourage diverse and stable filter representations. Another direction involves applying the framework to multi-center MRI datasets to better understand how acquisition variability influences degeneration and robustness.

Model compression techniques such as structured pruning, low-rank factorization, or filter-bank regularization might reduce the observed redundancy in deeper convolutional layers. A very promising direction will be extending the methodology to 3D CNNs or vision transformers since these architectures may exhibit considerably different degeneration patterns.

Future work should also include comprehensive robustness stress testing under clinically realistic perturbations, such as noise, motion, contrast shifts, and domain shifts.

Lastly, looking at degeneration dynamics in continual or incremental learning situations might help provide insights into how representations evolve or degrade as models experience newer data over time.

VIII. CONCLUSION

This study demonstrates how the *CNN Filter DB* analytical framework provides an effective and interpretable means of diagnosing degeneration within convolutional neural networks trained for brain tumor MRI classification. Although the ResNet-based models yield high predictive performance, filter-level analysis shows substantial structural degradation of filters, especially in deeper convolutional layers. Entropy collapse, rising sparsity, PCA basis contraction, and large KL-symmetric drifts collectively indicate that many filters become

structurally unstable, weakly informative, or redundant over the course of fine-tuning on limited medical imaging data.

These degeneration patterns are similar to the phenomena reported in the original *CNN Filter DB* study, and further demonstrate that high classification accuracy does not guarantee either robustness or representational reliability. The findings strongly suggest capacity control, data diversity, and degeneration-aware training strategies as key focus areas for safety-critical medical imaging applications.

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